



Spring MVC with Boot – WAR Model

- ♦ You can Develop as WAR file with Spring Boot
- ♦ WAR file is deployable application which can be deployed in any External Container to run.

How to deploy War file to Tomcat8

- 1) Make Sure that JDK8 is Installed.

```
E:\JDK1.8.0
E:\JDK1.8.0\bin
E:\JDK1.8.0\lib
```

- 2) Set JAVA_HOME

```
set JAVA_HOME=E:\JDK1.8.0
```

- 3) Make Sure that Tomcat8.5 is Installed.

```
E:\Tomcat8.5
E:\Tomcat8.5\bin\tomcat8
E:\Tomcat8.5\lib\*.jar
E:\Tomcat8.5\conf\server.xml
E:\Tomcat8.5\conf\web.xml
E:\Tomcat8.5\conf\tomcat-users.xml *
```

- 4) Update server.xml file

```
<Server port="21" shutdown="SHUTDOWN">

<Connector port="5050" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="5443" />
```

- 5) Make sure that Maven is Installed.

A) download the Maven from <https://maven.apache.org/download.cgi>
apache-maven-3.8.2-bin.zip

B) Extract the Zip and Copy to E:\ drive.

```
E:\apache-maven-3.8.2
E:\apache-maven-3.8.2\bin\mvn
E:\apache-maven-3.8.2\lib\*.jar
E:\apache-maven-3.8.2\confg\settings.xml
```



6) Set Maven Path

```
set path=%path%; E:\apache-maven-3.8.2\bin;
```

7) Add an user with roles manager-gui and manager-script in **tomcat-users.xml**

```
<user username="srinivas"  
      password="srinivas"  
      roles="manager-gui,manager-script" />
```

8) Add above Tomcat's user in the Maven setting file,
(**Maven will use this user to login Tomcat server**).

```
<server>  
<id>jlcTomcatServer</id>  
<username>srinivas</username>  
<password>srinivas</password>  
</server>
```

9) Add Tomcat7 Maven Plugin in pom.xml

```
<plugin>  
<groupId>org.apache.tomcat.maven</groupId>  
<artifactId>tomcat7-maven-plugin</artifactId>  
<version>2.2</version>  
<configuration>  
<url>http://localhost:5050/manager/text</url>  
<server>jlcTomcatServer</server>  
<path>/myjlcapp</path>  
</configuration>  
</plugin>
```

demo3-mvc-boot: Spring MVC with Boot (WAR)

Working Steps:

- 1) Copy **demo1-mvc-boot** as **demo3-mvc-boot**.
- 2) Update application.properties

No Properties Required

- 3) Update **MyBootApplication.java**



4) Update pom.xml

A) Add following 6 Dependencies

spring-boot-starter-web
jstl
standard
bootstrap
jquery
spring-boot-devtools

B) Add following 2 Plug-Ins

maven-compiler-plugin
tomcat7-maven-plugin

5) Open Command Line and Start the Tomcat8.5

E:\Tomcat8.5\bin>tomcat8

6) Open Another Command Line and run the Maven Command

...\demo3-mvc-boot>**mvn clean compile package tomcat7:deploy**

7) Open Browser and type <http://localhost:5050/myjlcapp>

Note : Maven Useful Commands

Deploy command => mvn tomcat7:deploy

UnDeploy command => mvn tomcat7:undeploy

ReDeploy command => mvn tomcat7:redploy

demo3-mvc-boot: Files Required

1. index.jsp	Same as Demo1
2. booksList.jsp	Same as Demo1
3. Book.java	Same as Demo1
4. IndexController.java	Same as Demo1
5. BooksController.java	Same as Demo1
6. JLCWebConfig.java	Same as Demo1
7. MyBootApplication.java	Updated in Demo3
8. application.properties	Updated in Demo3
9. pom.xml	Updated in Demo3



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7)MyBootApplication.java

```
package com.jlcindia.springboot;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.boot.builder.SpringApplicationBuilder;
import org.springframework.boot.web.servlet.support.SpringBootServletInitializer;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication extends SpringBootServletInitializer{

    public static void main(String[] args) {
        SpringApplication.run(MyBootApplication.class, args);
    }
    @Override
    protected SpringApplicationBuilder configure(SpringApplicationBuilder builder) {
        return builder.sources(MyBootApplication.class);
    }
}
```

8)application.properties

```
#No Properties Required
```



9)pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo3-mvc-boot</artifactId>

<version>1.0</version>
<packaging>war</packaging>
<name>demo3-mvc-boot</name>
<description>This is My Spring MVC App with Boot</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>jstl</groupId>
<artifactId>jstl</artifactId>
<version>1.2</version>
</dependency>

<dependency>
<groupId>taglibs</groupId>
<artifactId>standard</artifactId>
<version>1.1.2</version>
</dependency>

<dependency>
<groupId>org.webjars</groupId>
<artifactId>bootstrap</artifactId>
<version>4.3.1</version>
</dependency>
```



```
<dependency>
<groupId>org.webjars</groupId>
<artifactId>jquery</artifactId>
<version>3.4.1</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>
<build>
<finalName>myjlcapp</finalName>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>

<plugin>
<groupId>org.apache.tomcat.maven</groupId>
<artifactId>tomcat7-maven-plugin</artifactId>
<version>2.2</version>
<configuration>
<url>http://localhost:5050/manager/text</url>
<server>jlcTomcatServer</server>
<path>/myjlcapp</path>
<update>true</update>
</configuration>
</plugin>
</plugins>

</build>
</project>
```